

trading skills.md

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Summary

This document serves as a comprehensive **technical blueprint** for a sophisticated high-frequency trading system focused on **quantitative options analysis**. It bridges the gap between high-level financial research and practical software engineering by defining the specific **coding skills, mathematical models, and data infrastructure** required to execute complex strategies. The text details a **Master Composite Execution Matrix** that uses real-time indicators like implied volatility, gamma exposure, and open interest to generate automated **GO/NO-GO trading signals**. Ultimately, the source functions as a **rigorous implementation guide** that prioritizes ultra-low latency, strict risk management through emergency kill-switches, and a robust framework for backtesting market theories.

Keywords: Quantitative Options Trading, High-Frequency Programming, Data Engineering Infrastructure, Market Microstructure Analysis, Risk Management Systems

Content

SKILLS.md Executive Summary This document outlines the technical, quantitative, and soft skill requirements for a Quantitative Options/HFT Coding Agent and Live Trading Analyst. It also translates high-level research goals (e.g. Volatility-OI Nexus, Gamma Hedging, Lead-Lag signals) into concrete data-processing and algorithmic tasks. We list each required skill with proficiency level, typical responsibilities, example tools/libraries, and measurable acceptance criteria. We specify data inputs (tick-level, market data, option chains, etc.), sampling rates, latency targets, and storage needs for each feature. We present a Master Composite Execution Matrix (a decision table) that cross-references PCR regimes with IV trends, Gamma Exposure zones, and open-interest behavior, indicating clear GO/NO-GO triggers. Pseudocode snippets demonstrate key composite signal computations (Volatility-OI Nexus, Gamma hedging clues, Premium vs Spot divergence, Time-of-Day filters, No-Trade signatures). Finally, we define a unit/integration test checklist and outline

monitoring, alerting, and fail-safe procedures (circuit breakers, kill switches, degraded-mode) for live deployment. This SKILLS.md is concise, structured, and ready for a development team to implement, with references to industry standards and libraries.

Technical Skills Python (Expert) – Develop and prototype algorithms and data pipelines.

Tasks: Ingest and process tick/quote data, implement signal computations, build backtest frameworks and monitoring scripts.

Example Tools: Pandas, NumPy, TA-Lib, Numba for JIT optimization, Matplotlib/Plotly for visualization, PyTest for testing.

Acceptance Criteria: Clean, modular code with full test coverage; meeting latency targets (e.g. end-to-end signal update within ~100ms under load); proper handling of edge cases and missing data.

C/C++ (Expert) – Implement latency-critical components (data handlers, order routing).

Tasks: Build low-level data listeners (e.g. via sockets or pcap), optimized numeric kernels (e.g. option pricing, Greeks computation), ultra-low-latency order gateways.

Example Tools: GCC/Clang compilers, Boost.Asio for networking, Intel TBB/OpenMP for parallelism.

Acceptance Criteria: Achieve microsecond-level response (e.g. <5μs order generation), memory-safe code, benchmarked throughput, integration with exchange APIs, adherence to clean code standards.

Data Engineering & Databases (Advanced) – Manage large historical and streaming datasets.

Tasks: Design schemas for tick data and option chains; build ETL pipelines from exchanges/venues; optimize for query performance.

Example Tools: kdb+/q for time-series (millions of ticks/sec), InfluxDB/ClickHouse for aggregated metrics, Kafka/RabbitMQ for message queues.

Acceptance Criteria: Data ingestion at production rates without loss; sub-second queries on multi-day data; data compression/archival policies; clear documentation of data schemas.

Real-Time Messaging & APIs (Advanced) – Interface with market data and execution systems.

Tasks: Connect to exchange feeds (FIX, binary protocols), parse L1/L2 quotes and trades, send orders via FIX/REST/WebSocket APIs.

Example Tools: QuickFIX (C++/Python) for FIX, exchange SDKs (e.g. CME MD/AQ APIs), sockets/ZeroMQ for custom feeds.

Acceptance Criteria: Continuous data feed with <5ms end-to-end lag; automatically reconnect/handle dropouts; accurate message parsing and order acknowledgements; adhere to exchange protocols.

Distributed & Low-Latency Infrastructure (Intermediate) – Ensure fast, scalable execution.

Tasks: Deploy services on dedicated Linux servers (tuned network/stack), containerize (Docker), maintain host OS (kernel tuning, NIC settings).

Example Tools: Linux (Ubuntu/CentOS), Docker/Kubernetes, Linux perf/FlameGraphs, DPDK for ultra-low-latency networking.

Acceptance Criteria: Consistent throughput with <1% CPU overhead from OS, microsecond timestamp accuracy (e.g. via PTP or GPS), auto-scaling pipelines for load.

Version Control & CI/CD (Intermediate) – Manage code quality and deployment.
Tasks: Write unit/integration tests, use Git workflows, continuous integration (Jenkins/GitHub Actions), code reviews.

Acceptance Criteria: 100% of new code covered by tests; all merges pass linting and performance benchmarks; automated deployment scripts.

Trading & Risk Systems (Intermediate) – Integrate with risk monitoring tools.
Tasks: Query position/risk databases, enforce risk limits, log PnL and performance.

Example Tools: Custom in-house risk API, or industry platforms (Murex/Calypso), FIX order routing flags, FINRA/T+1 reports.

Acceptance Criteria: Real-time risk checks on each trade (no breaches allowed); compliance with exchange regulatory feeds; logging for audits.

Quantitative Skills
Options Pricing & Greeks (Expert) – Model option values, sensitivities and surfaces.
Tasks: Compute Black-Scholes/Merton Greeks ($\Delta, \Gamma, \Theta, \nu$); calibrate implied volatility (IV) surfaces (e.g. SVI/JW models); model time decay.

Example Tools: QuantLib or custom C/Python implementations; use SciPy for calibration.

Acceptance Criteria: Greeks consistent with exchange data (e.g. <1% error vs known values); IV surface smoothness (no arbitrage); correct θ acceleration curves; documented validation against benchmarks.

Implied Volatility and Skew Analysis (Advanced) – Analyze IV trends and skew behavior.
Tasks: Track IV of calls vs puts across strikes (IV skew); measure IV changes over time; detect “IV crush” or expansion events.

Acceptance Criteria: Automated computation of IV skew metrics each minute; statistical validation (e.g. correlation with realized vol); timely flags when IV trend diverges from norms.

Gamma Exposure (GEX) & Market Maker Dynamics (Advanced) – Compute net gamma and hedging signals.

Tasks: Aggregate per-strike gamma (Γ) weighted by open interest to get net Gamma Exposure; find “zero gamma” price levels; monitor delta-hedging triggers.

Example Tools: Python/NumPy for calculations, possibly accelerated by Numba/Cython; SpotGamma-like models for dealer vs customer flows.

Acceptance Criteria: GEX time series updating in real time (<1s lag) using live trades; correct identification of gamma flip zones; backtested ability to signal hedging events.

Order Flow & Open Interest Analytics (Expert) – Analyze volume, open interest, and positioning.
Tasks: Calculate Put/Call ratios (PCR) from live volume and OI; detect OI walls (strikes with large OI); compute Max Pain levels from OI distribution; track intraday OI changes.

Example Tools: Pandas for data tables, NumPy for numeric ops.

Acceptance Criteria: Correct PCR and OI metrics each minute; table of top OI strikes; real-time “smart money straddle shifts” via cumulative OI at multiple strikes; tests confirming data aggregation logic. Market Microstructure (Advanced) – Understand tick-level behavior and time-of-day effects.

Tasks: Analyze bid/ask book, trades, and price action patterns; segment day (open/midday/close) and compare behaviors; incorporate execution microstructure into signals.

Acceptance Criteria: Verified detection of typical patterns (opening auction spikes, lunchtime lulls, end-of-day runs); empirically-backed time-of-day filters (e.g. avoid trading during midday with low volume). Statistical & Machine Learning (Intermediate) – Analyze multi-variate signals and implement algorithms.

Tasks: Correlation analysis (e.g. PCR vs IV changes), regression/pca for feature reduction, anomaly detection on feed quality. May use ML models (scikit-learn, XGBoost) for regime classification.

Example Tools: scikit-learn, PyTorch/TensorFlow for custom models.

Acceptance Criteria: Model validation metrics (accuracy/ R^2) on historical signals; automated feature pipelines with unit tests; resource usage within limits. Signal Development & Backtesting (Advanced) – Build and test trading signals.

Tasks: Implement trading logic combining features (Vol-OI composite, gamma hedging clues, etc.); backtest on historical intraday data; refine signals.

Example Tools: Backtesting frameworks (e.g. Zipline, QSTrader) or custom engine; vectorized Pandas calculations.

Acceptance Criteria: Backtest results with realistic costs/slippage; Sharpe ratios and drawdowns meeting design criteria; reproducible performance in paper trading before live. Soft Skills Decision-Making Under Pressure – Rapid judgment on GO/NO-GO signals.

Tasks: Assess real-time signal validity, weigh risk-reward quickly, collaborate with trading desk in volatile markets.

Acceptance Criteria: Consistent justification of actions; post-mortem analysis of decisions; positive peer feedback on sound reasoning. Incident Response and Risk Awareness – React to system anomalies or market crises.

Tasks: Execute kill-switch protocols; communicate with risk managers; adapt strategies during market dislocations (circuit breaker events).

Acceptance Criteria: Adherence to fail-safe checklists; timely escalation of alerts; no major errors in simulated crash drills. Communication & Collaboration – Work with developers, traders, and compliance.

Tasks: Document algorithms and tests; present signal rationale; write clear code comments.

Acceptance Criteria: Reviewed documentation for each signal; code reviewed and approved; training materials for other analysts. Adaptability & Learning – Stay updated on market structure changes.

Tasks: Research new order types, exchange features, regulatory updates; refine models accordingly.

Acceptance Criteria: Incorporate at least one new data source or technique per quarter; continuous improvement of models.

Feature Engineering & Signal Computation

PCR and Sentiment Signals: Compute real-time **Put/Call Ratio** (e.g. volume of puts / volume of calls, or vice versa) for index options. Categorize into regimes (e.g. PCR <0.7, 0.7–1.0, >1.0). Tasks: stream aggregated volumes from trades or quotes; update ratio each second. Feature: **PCR Trend** (derivative) and **PCR Change from Open** (difference from session start). Input: tick/trade data (calls and puts) at 1–5 second sampling. Latency: <1s for updates. Storage: keep 1-minute PCR history; raw tick data stored (unspecified retention). Acceptance: ratio matches known values (e.g. CBOE data).

Implied Volatility Skew & Theta: From live option quotes (or trades), extract at-the-money (ATM) and near-the-money (NTM) IV for calls and puts. Compute IV Skew = CE_IV / PE_IV or difference at given strike. Track IV Trend: e.g. moving averages of IV or its slope over 5–15 min. Compute **Time Decay (Theta)**: using option pricing (QuantLib/BS) to estimate acceleration of theta for NTM options. Input: continuous quote stream at 1s or tick resolution. Output: skew and theta curves per minute. Acceptance: stable IV surface with correct sign (calls typically > puts if bullish skew).

Gamma Exposure (GEX): Calculate net gamma = $\sum (\Gamma_i * \text{net open interest}_i)$ across strikes and expirations. Use intraday OI estimates (adjust official OI with trade flow). Identify **Zero-Gamma level**: price where net gamma ≈ 0 . Determine **GEX Zones**: positive (>0) or negative (<0) net gamma. Input: trade prints, intraday OI, option Greeks; sampling: update on every sizable trade or max 1s. Latency: <100ms for gamma calc. Acceptance: GEX sign flips match major price moves in test data; zero-gamma aligns with support/resistance from backtest.

OI Walls & Institutional Positioning: Detect **OI Walls** by finding strikes with unusually large open interest (e.g. top 5% of OI values). Track **Total OI** and **Intraday OI Changes** per strike. Compute **Max Pain** (price minimizing intrinsic value sum) each interval and shifts. Compute **Cumulative OI shifts** (e.g. sum of changes across straddle/strangle strikes). Input: official OI (updated EOD) plus intraday volume adjustments; sampling: OI updates as received (min once/day, intraday adjustment every 1–5min). Latency: intraday OI used within 1min of trade. Acceptance: stress-tested on sample sessions with known OI walls; max-pain computation sanity-checked.

Volume-Premium VWAP: For NTM call and put series, compute **Premium Volume Weighted Average Price** ($PVWAP = \sum(\text{price} * \text{volume}) / \sum(\text{volume})$) and underlying **Spot VWAP** over rolling windows (e.g. 5min). Monitor deviation: e.g. ($PVWAP_{\text{call}} - \text{SpotVWAP}$). Input: trade ticks for underlying and selected options; sampling tick-level or aggregated 1s. Acceptance: VWAPs converge to expected under neutral conditions; divergence correlates with directional moves (validated on backtest).

Composite Signal Computations: Combine features as per research objectives. For example, **Volatility-OI Nexus**: when (PCR

in bull regime AND IV Skew collapsing AND OI on calls spiking) $\Rightarrow$ signal a “Vol-Crush Bull Trap” (NO-GO on call buys). Conversely, (PCR bull regime AND IV expanding AND increasing call OI) $\Rightarrow$ “High-Velocity Breakout” (GO). **Gamma Hedging Clue:** near zero-gamma price + heavy OI wall in calls + accelerating underlying = trigger for market-maker shorting (GO to buy calls anticipating squeeze). **Lead-Lag (Premium vs Spot):** if spot range-bound, but NTM call premium breaks above VWAP while PCR rising and ATM call OI dropping (short-covering), then this signals latent bullish accumulation (GO). Each composite requires synchronized input streams (PCR, IV, OI, spot).

Time-of-Day Segmentation: Tag data into three windows: Morning Open (first 60m), Midday Dull (lunch hours), Afternoon Run (last hour). Adjust filters per block: e.g. ignore small signals midday, tighten criteria at open, relax on final hour. Monitor PCR change, Max Pain shift, OI flows in context. Acceptance: backtests show significantly different performance by time block; rules flag historically correct/no-trade periods.

No-Trade (Trap) Signature: Detect when **PCR is flat (near neutral) AND Max Pain stationary AND massive OI build-up on both CE and PE** (e.g. writes of short strangles/straddles). This indicates premiums draining to theta – a signal to avoid new longs. Implementation: if flat PCR (e.g. within $\pm 5\%$ of open) and Net OI change $> X$ at adjacent strikes with balanced CE/PE buildup, raise NO-TRADE flag.

Input Data Types: Level-1 market data (best bid/ask, last trade) for underlying index; Level-1/2 option quotes and trades; daily OI snapshots (exchange) + intraday trade flow; historical tick data for backtests.

Sampling Rates: Underlying and option quotes at millisecond-level or 1-second bars; aggregated features (PCR, GEX) updated 1–5s. **Latency Targets:** Final composite signals computed within $\sim 100\text{ms}$ of data ingestion (preferably $< 50\text{ms}$) to allow timely execution. **Storage/Retention:** Raw tick data stored long-term (e.g. 1 year or more); aggregated features (per-minute summaries) kept for multi-year backtesting. (Where unspecified, marked as “unspecified”).

Master Composite Execution Matrix

Notes: PCR regimes refer to put/call flow (e.g. $\text{PCR} \ll 0.7$ indicates call-heavy bullish sentiment). “Expanding IV” means rising implied volatility (momentum likely); “Crushing IV” is a rapid IV drop (trap risk). Positive GEX (net dealer gamma long) implies range-biased market; negative GEX implies trend bias. OI behavior distinguishes accumulation (e.g. build-up in calls) vs balanced writing. Triggers: GO when all bullish or bearish signals align (e.g. bull PCR + rising IV + heavy call OI; or bear PCR + rising IV + heavy put OI). NO-GO when mixed signals (e.g. bullish PCR but IV collapsing) indicate traps. The neutral regime requires further confirmation. This matrix guides automated decision logic for each minute.

Algorithmic Pseudocode Snippets

Test & Validation Checklist

Unit Tests for Feature Calculations: e.g. feed synthetic option chains to verify IV skew, Gamma, Theta, and PCR calculations produce expected results.

Integration Tests for Data Pipelines: simulate streaming trade data and ensure the end-to-end flow (ingest $\rightarrow$ compute features $\rightarrow$ trigger signals) works correctly.

Backtesting Tests: apply

signals on historical intraday sessions; verify results against known outcomes (e.g. bull-trap episodes should have given NO_TRADE).

Latency and Throughput Tests: benchmark data ingestion and signal update times under load; ensure single-update (from data to alert) completes <100ms.

Edge Case Handling: test market halts, data gaps, zero volume periods; verify safe behavior (e.g. no false signals on missing data).

Performance Regression: monitor memory/CPU use; integrate checks that new code does not degrade performance beyond X%.

Deployment Dry-Run: simulate failover and kill-switch activation to ensure automated processes intervene as designed.

Monitoring, Alerting, and Fail-Safes

Automated Strategy Monitoring: Continuously track key metrics per strategy (order rate, cancel/fill ratio, position P&L, net delta). Use a monitoring platform (e.g. Grafana or 3rd-party) to visualize these in real time. Set thresholds (e.g. abnormal order rate or slippage) to trigger immediate alerts.

Circuit Breakers & Kill Switch: Implement an emergency kill-switch that, when activated, **immediately** disables all trading and cancels working orders. Trigger this on extreme losses, regulatory halts, or rule violations. For example, PnL drawdown > X% or catastrophic system error should invoke kill-switch.

Degraded-Mode Behavior: Define fallback procedures if market data or connectivity fails (e.g. switch to static spreads or close positions). For instance, if option feed stalls >5s, shut off new orders and clear near-zero gamma positions to minimize exposure.

Alerting: Send automated alerts (email/SMS/Slack) to analysts and risk managers on critical events: signal breaches, latencies > threshold, kill-switch events, etc. Maintain an audit log with microsecond timestamps for all signals and actions for compliance.

Risk Limits Enforcement: Monitor positions against pre-set limits (size, delta, gamma). If any limit is breached or about to, suspend trading until reviewed. This is in addition to kill-switch and serves as a pre-trade check.

Audit & Compliance: Ensure every kill-switch activation and signal is logged with context. Provide operators with dashboards summarizing system health and strategy activity (helps meet regulatory requirements).